J. Mol. Biol. (1988) **202**, 245–253

Characterization of the Termination Phenotypes of Rifampicin-resistant Mutants

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(Received 4 September 1987, and in revised form 15 February 1988)

Rifampicin-resistant (Rif^r) mutations map in the *rpoB* gene encoding the β subunit of *Escherichia coli* RNA polymerase. We have examined the effect of each of the 17 sequenced Rif^r mutations in our collection on transcription termination. The effect of each Rif^r mutation was measured at three types of terminators: simple terminators requiring only RNA polymerase to terminate *in vitro*, and complex terminators requiring either Rho or Tau for *in-vitro* termination. Almost every Rif^r allele examined (14/17) affected read-through at one or more of these terminators. We found that mutations with similar termination phenotypes were clustered suggesting functional specialization within the region of *rpoB* defined by the Rif^r mutations. The interaction of the Rif^r mutations with the defective *rho15* allele was also investigated. Only two Rif^r mutations suppress the termination defect of *rho15* strains. We discuss models to explain how this region of the β polypeptide might be involved in the process of transcription termination.

1. Introduction

RNA polymerase in *Escherichia coli* plays an integral role in the process of transcription termination. It responds to the signals in the nascent transcript that interrupt the process of transcription elongation and interacts with auxiliary protein factors involved in termination (Adhya & Gottesman, 1978; Rosenberg & Court, 1979; Platt, 1986). During elongation, core RNA polymerase (and associated proteins) moves along the DNA template, creating a locally melted bubble of DNA. Nucleotide addition occurs at or near the leading edge of the transcription bubble and the nascent RNA transcript interacts with about 12 bases of the melted DNA to form a DNA–RNA hybrid (Yager & von Hippel, 1987). Elongation is processive and continues until a pause site is reached. Termination involves the following steps: recognition of a pause site by RNA polymerase, destabilization of the nascent RNA–DNA hybrid, closure of the the DNA bubble, and release of both transcript and RNA polymerase from the DNA template.

Termination sites have been extensively investigated. Terminators are categorized either as simple (factor-independent) or complex (factor-dependent) (Yager & von Hippel, 1987). A simple terminator consists of a G+C-rich hairpin loop followed by five or more U residues and allows efficient termination in *in-vitro* transcription systems containing only template and RNA polymerase. The hairpin loop signals a transcription pause site while the unstable dA·rU base-pairs promote dissociation of the nascent RNA–DNA hybrid (Farnham & Platt, 1980; Martin & Tinoco, 1980). A complex terminator contains a pause signal but requires interaction with additional proteins such as Rho, Tau or NusA to complete the termination process. The role of Rho protein may be analogous to the dA·rU base-pairs; it acts as a helicase to dissociate the paused nascent RNA–DNA hybrid (Brennan *et al.*, 1987). The role of other proteins in promoting termination has yet to be determined.

In contrast, little is known concerning the role of RNA polymerase in termination. One approach to this problem, taken by Robert Glass and his colleagues, has been to make amber mutations throughout *rpoB*, the gene encoding the β subunit of RNA polymerase, and to analyze the mutant phenotypes following suppression with a variety of nonsense suppressors (Nene & Glass, 1984). They have assayed the ability of the mutants to plate λ phage that are unable to grow on normal host cells because they lack the N-mediated antitermination system required for lytic growth. Using this indirect assay, they identified mutations in several regions of *rpoB* that may be involved in termination (Nene

0022–2836/88/140245–09 \$03.00/0

& Glass, 1984). An alternative approach described here is to perform an extensive analysis of a class of mutants, the Rif^r† mutants, believed to have altered transcription termination phenotypes. Some Rif^r mutants suppress the termination defect of mutant *rho* alleles. The *rpoB203* mutation (Guarente & Beckwith, 1978) restores normal termination to strains containing several different defective *rho* alleles, including a *rho* amber allele, while the *rpoB101* mutation was shown to restore normal termination to strains containing *rho15* (Das *et al.*, 1978). Other Rif^r mutations have been shown to alter termination at specific terminators. The work of Neff & Chamberlin (1980) was the first to show that *in-vitro* termination is altered by some Rif^r mutations. Although RNA polymerase from wild-type cells was unable to terminate *in vitro* at the bacteriophage T_3 early terminator T_3T_e, RNA polymerase from two Rif^r mutants was able to do so. In addition, RNA polymerase from several Rif^r mutants isolated by Yanofsky & Horn (1981) with altered *trp* attenuation *in vivo* proved to have altered termination properties *in vitro* (Fisher & Yanofsky, 1983).

To date, no systematic study had been done to determine which Rif^r mutations alter the termination properties of RNA polymerase, and whether mutations with similar effects are clustered within *rpoB*.

We have a collection of Rif^r mutations that includes 17 different mutations affecting 14 amino acids. These alleles are located in three clusters in the central region of the β polypeptide (Jin & Gross, 1988, and Fig. 2.). Cluster I includes amino acid residues 507 to 533 and contains 13 of the Rif^r mutations. Cluster II includes amino acid residues 563 to 572 and contains 3 of the Rif^r mutations. Cluster III is defined by one Rif^r affecting amino acid residue 687. We used this collection of Rif^r mutations to assess the involvement of this region of the β subunit in the termination process. We describe the effect of each Rif^r mutation on the efficiency of termination *in vivo* at simple and complex terminators, and on the termination properties of strains containing the defective *rho15* allele.

2. Materials and Methods

(a) *Bacterial strains*

The Rif^r mutants used in this study are presented in Table 1 and have been described (Jin & Gross, 1988). The effects of these Rif^r alleles on termination at simple and complex terminators were determined in Rif^r derivatives of CAG8102, a *strA*$^+$ derivative of K37.

The effects of the Rif^r mutations on the *rho15* allele were assayed in CAG3284, a *gal-3* derivative of MG1655. The Rif^r alleles were introduced into CAG3284 by cotransduction with a linked Tn*10* marker as described

Table 1
Strains used in this study

Strain	Relevant genotype	Source/ reference
MG1655	*E. coli* K12 wild-type	CGSC
CAG3284	MG1655 *gal-3*	This work
K37	*galK2 rpsL200* λ^-	CGSC
CAG8102	*rpsL*$^+$ derivative of K37	M. Singer
CAG3807	MG1655 *gal-3 rho15 ilv*::Tn*5* (linked to *rho15*)	This work

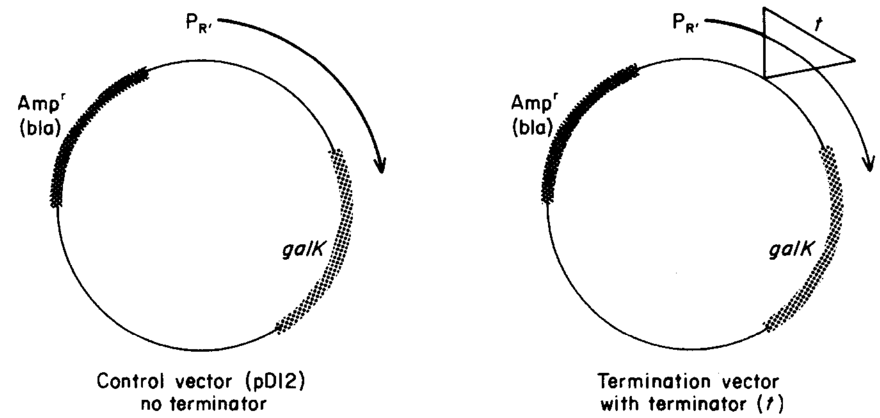
Rif^r allele (*rpoB*)	Amino acid residue affected	Amino acid change
Cluster I		
3445	Δ(507-511)	ΔGly Ser Ser Gln Leu and inserts Val
101	513	Gln to Leu
8	513	Gln to Pro
113	516	Asp to Asn
148	516	Asp to Val
3051	517	Insert Gln and Asp
3595	522	Ser to Phe
2	526	His to Tyr
3401	529	Arg to Cys
3402	529	Arg to Ser
114	531	Ser to Phe
3449	Δ532	ΔAla
3443	533	Leu to Pro
Cluster II		
3370	563	Thr to Pro
111	564	Pro to Leu
7	572	Ilv to Phe
Cluster III		
3406	687	Arg to His

(Jin & Gross, 1988). *rho15* was transduced into the Rif^r derivatives of CAG3284 by selection for a linked Tn*5* at the *ilv* locus. The presence of the *rho15* allele was verified by its temperature sensitive (Ts) growth phenotype where possible. When the Rif^r alleles themselves conferred a Ts phenotype, the presence of *rho15* was verified by transduction to the wild-type strain. It is necessary to transduce *rho15* into the Rif^r strains, rather than the other way around, because *rho15* strains have a recombination deficiency (S. Adhya, personal communication).

(b) *General bacterial techniques and enzyme assays*

General bacterial techniques were as described (Jin & Gross, 1988). Galactokinase (McKenney *et al.*, 1981) and β-lactamase (Tomizawa, 1985) activity are expressed as a differential rate of synthesis. The differential rate of synthesis is the slope of the line generated when enzyme activity is plotted *versus* cell growth (A_{450}). For measuring the termination phenotypes of Rif^r mutants at different terminators, cultures were grown at 37 °C in M9-glucose supplemented with all amino acids, vitamins, nucleosides and ampicillin (50 μg/ml). For measuring the effect of Rif^r mutants on *rho15*, cultures were grown at 32 °C in M9-fructose supplemented with all amino acids, vitamins, nucleosides and 10 mM-D-fucose. Each strain was sampled 3 to 4 times within 2 doublings during exponential growth. β-Lactamase assays were performed on the same extracts used for the galactokinase assays.

† Abbreviations used; Rif^r, rifampicin-resistant; Ts, temperature-sensitive; galK, galactokinase.



Termination-assaying vectors

Plasmid	Terminator	Type of terminator	%RT† in CAG8102 ($rpoB^+$)
pD12	none	—	100
pDS34	λtL_{2a}	Simple	17
pDS367	λtL_{2b}	Simple	27
pQS81	$t\lambda'_{R2}$	Rho-dependent	50
pSG64	$\lambda t'_{J4}$	Rho-dependent	10
pD129	λt_{R1}	Rho-dependent	7

† % RT is defined in Materials and Methods. Numbers are the average of 3 independent determinations of differential rate of synthesis as described in Materials and Methods. The values of the 3 determinations generally deviated less than 10% from the average value.

Figure 1. A schematic description of the vectors to assay the termination phenotype. Amp, ampicillin; bla, β-lactamase.

D-[1-^{14}C]galactose (40 to 50 mCi/mmol) was purchased from Research Products International Corp. (Mount Prospect, IL). Nitrocephin was from BBL Microbiology Systems (Cockeysville, MD).

(c) *Terminator vectors*

The vectors used to assay the termination phenotypes of the Rif^r mutations were constructed and characterized by Luk & Szybalski, (1982, 1983) and Szybalski *et al.* (1983) and are described schematically in Fig. 1. In each case, a particular λ terminator is inserted between the promoter $\lambda P_{R'}$ and the galactokinase (*galK*) gene. We chose 5 terminators with different efficiencies for this study. Three of these terminators are dependent on Rho for termination, since their efficiency was decreased 4 to 8-fold in a *rho15* strain background *in vivo* and they require Rho to terminate at these terminators *in vitro* (Luk & Szybalski, 1982; Luk, 1983; Szybalski *et al.*, 1983). The other 2 are simple terminators with a characteristic G+C-rich stem and loop structure followed by a string of U residues. RNA polymerase is able to terminate at these terminators *in vitro* in the absence of auxiliary factors (Luk & Szybalski, 1983).

(d) *Procedure for determining termination phenotype*

A simple comparison of galactokinase (galK) activity in wild-type and mutant cells containing a termination vector may not accurately determine a difference in termination efficiency. If the *rpoB* mutation affects promoter strength or plasmid copy number, this will influence the galK value. To obtain an accurate estimate of termination, the ratio of galK activity to β-lactamase (bla) activity (from the *bla* gene on the same plasmid) in cells carrying a termination vector is normalized to the galK/bla ratio in cells of the same strain that carries the vector without a terminator. This determines the fraction of transcripts that read-through a particular terminator (%RT), in a particular strain, independent of promoter strength and plasmid copy number. The "termination phenotype" of a Rif^r mutant is then defined as the relative RT; this is, the %RT of a terminator in the Rif^r mutant divided by the %RT in the wild-type strain. The procedure is as follows.

(1) Determine the differential rate of galK and bla synthesis for the wild-type and each Rif^r strain containing either the control plasmid or the terminator vectors. Each differential rate of synthesis is determined from the slope of the enzyme activity (galK or bla) *versus* A_{450} and is based on 3 to 4 data points. For each Rif^r mutant, the entire set of terminator plasmids as well as the control plasmid (no terminator) were assayed together. Each experiment included a wild-type control with its entire set of plasmids.

(2) Calculate the %RT for each Rif^r mutant at a terminator (t) by comparing the galK/bla ratio of the terminator vector with that of the control vector:

$$\%RT \text{ of a Rif}^r \text{ mutant at terminator } (t) = \frac{\text{galK/bla of terminator } (t) \text{ vector}}{\text{galK/bla of control vector}} \times 100.$$

(3) Calculate the "termination phenotype" or relative RT for each Rif^r mutant at terminator (t) by comparing

Table 2
Example of data used to determine termination phenotypes of Rif^r mutants

Termination-assaying vector	rpoB allele		
	rpoB$^+$	*rpoB2*	*rpoB8*
Control vector (no terminator)			
galK†	35·0	42·5	105·0
bla‡	0·62	0·37	1·48
galK/bla	56·5	114·8	70·9
λt_{L2a}			
galK	4·5	16·0	3·4
bla	0·47	0·23	0·90
galK/bla	9·6	69·6	3·8
%*RT*§	17	60	5
Relative *RT*‖	1·0	3·5	0·3
λt_{L2b}			
galK	12·5	15·0	17·0
bla	0·82	0·35	1·50
galK/bla	15·2	42·9	11·3
%*RT*	27	37	16
Relative *RT*	1·0	1·4	0·6
$\lambda t'_{R2}$			
galK	23·5	16·0	35·5
bla	0·83	0·23	1·80
galK/bla	28·3	69·6	19·7
%*RT*	50	61	28
Relative *RT*	1·0	1·2	0·6
$\lambda t'_{J4}$			
galK	3·0	7·5	8·0
bla	0·56	0·40	1·50
galK/bla	5·4	18·8	5·3
%*RT*	10	16	8
Relative *RT*	1·0	1·6	0·8
λt_{R1}			
galK	2·2	4·0	3·5
bla	0·60	0·32	1·50
galK/bla	3·7	12·5	2·3
%*RT*	7	11	3
Relative *RT*	1·0	1·6	0·4

† This value represents the differential rate of synthesis of galactokinase determined from the slope of enzyme activity *versus* A_{450}.

‡ This value represents the differential rate of synthesis of β-lactamase determined from the slope of enzyme activity *versus* A_{450}.

§ $\%RT = \dfrac{galK/bla \text{ of terminator } (t) \text{ vector}}{galK/bla \text{ of control vector}} \times 100.$

‖ $\text{Relative } RT = \dfrac{\%RT \text{ of Rif}^r \text{ mutant}}{\%RT \text{ of } rpoB^+}.$

For details, see Materials and Methods.

the %*RT* of the Rif^r mutant to that of the *rpoB*$^+$ parental strain:

$$\text{“Termination phenotype” at terminator } (t) \text{ or relative } RT = \frac{\%RT \text{ of the mutant}}{\%RT \text{ of the } rpoB^+ \text{ strain}}.$$

The termination phenotypes presented in this paper are based on 2 independent determinations of the differential rate of synthesis of galK and bla. The independent determinations of the termination phenotype were in close agreement. The termination phenotypes of *rpoB2*, *rpoB7* and *rpoB8* have been measured (Yanofsky & Horn, 1981). Our results are comparable to theirs (see Tables 3 and 5). Sample data for the wild-type strain and 2 mutant strains are presented in Table 2. The entire data set upon which our conclusions are based is available upon request.

3. Results

(a) *Termination phenotypes of Rif^r mutants at several simple and Rho-dependent terminators*

We examined the effects of the Rif^r mutations on termination at two simple terminators and three Rho-dependent terminators using a set of terminator plasmids derived from pBR322 constructed by Luk & Szybalski (1982, 1983) and Szybalski *et al.* (1983). Each plasmid contains a different λ terminator inserted between the $\lambda P_{R'}$ promoter and the *galK* gene (see Fig. 1). To determine the percentage of transcripts which read through a particular terminator (%*RT*), we normalized the galK/bla ratio from the plasmid containing the terminator to the galK/bla ratio from the plasmid with no terminator, as described in Materials and Methods. This corrects for effects of the *rpoB* mutation on promoter strength and plasmid copy number. The "termination phenotype" of a Rif^r mutant is defined as the relative *RT*; this is the %*RT* of a terminator in the Rif^r mutant divided by the %*RT* in the wild-type strain.

The termination phenotypes of the Rif^r mutants are reported in Table 3. Of the 17 Rif^r alleles we examined, 14 affected read-through $\geq$ twofold for at least one of the terminators assayed. It is possible that the three Rif^r mutants that did not affect read-through at these terminators would have termination phenotypes at other terminators. Seven Rif^r mutants (*rpoB3595*, *rpoB2*, *rpoB3401*, *rpoB3402*, *rpoB3370*, *rpoB111* and *rpoB3406*) increased read-through of at least one terminator placed upstream from the galactokinase structural gene. An equal number of mutants (*rpoB3445*, *rpoB101*, *rpoB8*, *rpoB113*, *rpoB148*, *rpoB3051* and *rpoB7*) decreased read-through of at least one terminator. A majority of the Rif^r alleles affected read-through at more than one terminator. Eight showed $\geq$ twofold and 12 showed $\geq$ 1·5-fold effects on read-through at two or more terminators. In every case where more than one terminator was affected, the direction of the effect at each terminator was the same.

There is a strong correlation between the location of the Rif^r mutation and its effect on termination. The Rif^r mutations are divisible into three clusters, of which the first two are located close together (Jin & Gross, 1988; Fig. 2). The first six mutations in cluster I and the most distal mutation in cluster II all decreased read-through (increased termination). The next four mutations in cluster I and the first two mutations in cluster II all increased read-through (decreased termination). This correlation suggests that there is functional specialization within the portion of β subunit identified by the

Table 3
Relative read-through of Rif^r mutants at different terminators

rpoB	Amino acid affected	Simple terminators		Rho-dependent terminators		
		λt_{L2a}	λt_{L2b}	$\lambda t'_{R2}$	$\lambda t'_{J4}$	λt_{R1}
Wild type	None	1·0	1·0	1·0	1·0	1·0
Cluster I						
3445	Δ(507-511)	0·9	0·8	**0·4**	1·0	0·9
101	513	**0·5**	0·8	0·7	**0·5**	**0·4**
8	513	**0·3**	0·6	0·6	0·8	**0·4**
113	516	1·0	1·0	0·9	1·0	**0·3**
148	516	1·0	0·9	0·9	**0·3**	0·6
3051	517	0·8	0·9	0·9	**0·5**	**0·5**
3595	522	**3·5**	**2·5**	**2·0**	**2·7**	**2·5**
2	526	**3·3**	1·3	1·1	1·6	1·5
3401	529	1·0	1·1	1·5	**6·0**	**7·5**
3402	529	0·8	0·9	1·1	**2·2**	**2·6**
114	531	0·9	0·9	1·1	0·9	0·9
3449	Δ532	1·0	1·0	1·0	1·0	0·9
3443	533	1·1	0·9	0·9	1·0	1·0
Cluster II						
3370	563	1·5	1·7	1·5	**2·1**	1·9
111	564	**2·5**	1·5	1·1	1·4	1·2
7	572	**0·5**	**0·5**	0·7	0·7	0·7
Cluster III						
3406	687	1·0	1·0	1·0	**2·0**	**6·2**

Relative *RT* is the %*RT* at a particular terminator by a Rif^r mutant normalized to the %*RT* at the same terminator in the wild-type parental strain (see Materials and Methods for a more detailed description). Values ≥2·0-fold are shown in bold face; values ≥1·5-fold are underlined. Numbers are the average of 2 independent determinations of the differential rate of synthesis as described in Materials and Methods. The values of the 2 determinations were in close agreement.

first two clusters of Rif^r mutations. Alterations in the N-terminal and most C-terminal portions of this region lead to more efficient termination while those in the middle result in less efficient termination.

Each Rif^r allele affects its own spectrum of terminators. Six of the Rif^r mutants affected read-through at both simple and Rho-dependent terminators; seven affected Rho-dependent terminators only; and one affected simple terminators only (Tables 3 and 5). Within this grouping each Rif^r mutant altered read-through at a variable number of terminators. For example, of those Rif^r mutants altering termination at Rho-dependent terminators only, two affected one terminator, four affected two terminators and one affected all terminators assayed (Table 5). In general, mutations located close together tended to affect similar terminators indicating that their mutational alterations are likely to have similar effects on RNA polymerase function.

(b) *Termination phenotypes of Rif^r mutants at a Tau-dependent terminator*

Terminator T_3T_e limits transcription to the early region of bacteriophage T_3 and is a very efficient Rho-independent terminator *in vivo*. T_3T_e functions poorly *in vitro* whether or not Rho is present (Neff & Chamberlin, 1980). Briat & Chamberlin (1984) purified a factor required for termination at T_3T_e *in vitro* called Tau. We used a terminator vector in which T_3T_e is inserted between promoter P_{gal} and the galactokinase structural gene (Chamberlin *et al.*, 1986) to examine the effect of Rif^r mutants on this terminator. Production of galactokinase as judged by enzyme assay or phenotype on MacConkey galactose plates was comparable in mutant and wild-type cells (data not shown). None of the Rif^r mutants appeared altered in termination at the T_3T_e terminator *in vivo*.

(c) *Effects of the Rif^r mutants on the termination defect of* rho15 *strains*

When an IS*2* element is inserted between P_{gal} and the galactose structural genes (*gal-3* mutation), the strong Rho-dependent terminator in IS*2* prevents transcripts from reaching the structural genes. As a consequence, cells appear white on MacConkey galactose indicator plates and are unable to grow on galactose. The defective Rho protein encoded by the *rho15* allele allows transcription to proceed through IS*2* into the *gal* structural genes, resulting in a Gal^+ phenotype. The Rif^r mutation *rpoB101* isolated by Das *et al.* (1978) restores termination and the Gal^- phenotype to a strain containing *gal-3 rho15*.

We wanted to test the ability of each of the Rif^r mutations to revert the termination defect of the *rho15* allele, and to correlate this phenotype with the effects on the terminators described above. We transduced the *rho15* allele into a strain background

Table 4
Interaction between Rif^r mutations and rho15

rpoB allele	galK expression† in *rho$^+$ gal$^+$* (A)	in *rho15 gal-3* (B)	Read-through‡	Read-through relative to *rpoB$^+$ rho15 gal-3*§	Colony color (*rho15 gal-3*) on MacConkey gal plates
rpoB$^+$	33	17	0·5	1·0	Red
Cluster I					
3445	16	10	0·6	1·2	Red
101	24	2	0·1	0·2	White
8	17	13	0·8	1·6	Red
113	25	11	0·4	0·8	Red
148	30	15	0·5	1·0	Red
3051	29	15	0·5	1·0	Red
3595	18	16	0·9	1·8	Red
2	28	‖	-	—	
3401	21	21	1·0	2·0	Red
3402	21	19	0·9	1·8	Red
114	18	16	0·9	1·8	Red
3449	19	15	0·8	1·6	Red
3443	18	14	0·8	1·6	Red
Cluster II					
3370	26	3	0·1	0·2	White
111	18	14	0·8	1·6	Red
7	25	13	0·5	1·0	Red
Cluster III					
3406	25	16	0·6	1·2	Red

† Cultures were grown at 32°C and induced with D-fucose. Numbers are the average of 2 independent determinations of the differential rate of synthesis as described in Materials and Methods. The values of the 2 determinations generally deviated less than 10% from the average value.

‡ *galK* expression in each *rho15 gal-3* (IS2) derivative (B) is expressed as a fraction of *galK* expression in the isogenic *rho$^+$ gal$^+$* strain (A). This gives read-through of the *gal-3* (IS2) terminator, as a fraction of "read-through" in the absence of a terminator. With *rho$^+$* and the *gal-3* IS2 insertion, *galK* expression is indistinguishable from background (regardless of the *rpoB* allele).

§ The read-through values for each Rif^r allele in *rho15 gal-3* are normalized to the read-through for *rpoB$^+$* in *rho15 gal-3*, to show the effect of the Rif^r allele relative to *rpoB$^+$*.

‖ Incompatible with *rho15*.

containing each of the Rif^r alleles and the *gal-3* mutation. We scored the Gal phenotype of the double mutants by colony color on MacConkey galactose plates and by galactokinase assays. These assays indicated that *rpoB101* and *rpoB3370* are the only Rif^r mutants that significantly increase termination at IS2 in a *rho15* background (Table 4). Eight of the Rif^r mutants decreased termination 50% or more. Because expression of *galK* is not detectable above background in the isogenic *rho$^+$* strain background we were unable to assess the affects of these Rif^r mutations on read-through at the IS2 terminator in strains containing the wild-type Rho protein. Thus, we do not know whether the increased read-through in these eight mutants is related to changes in recognition of the terminator or the mutant Rho15 protein.

During construction of these strains we discovered that the *rpoB2* allele was incompatible with *rho15*. We were unsuccessful in recovering the double mutant at temperatures ranging from 25 to 40°C. The basis of this incompatibility is unknown, but may be related to altered termination in the *rpoB2 rho15* strain background.

4. Discussion

We have investigated the termination phenotype of 17 Rif^r mutations by examining their effect on read-through at several terminators and on termination in strains with the defective *rho15* allele. The termination phenotypes of each allele are summarized in Table 5. Only two of the Rif^r mutations suppress the termination defect of *rho15*. In contrast, most of the Rif^r alleles (14/17) were found to alter termination efficiency at one or more terminators. In fact, the correspondence between Rif^r and altered termination is complete. The three alleles that do not affect termination in these assays have been found to affect antitermination (D. J. Jin & C. A. Gross, unpublished results).

These results indicate that the Rif^r mutations identify a region of the β subunit of RNA polymerase that is involved in termination. The grouping of alleles with similar termination phenotypes within the Rif^r region of *rpoB* encompassing clusters I and II (see Fig. 2) suggests further functional specialization. Rif^r mutations in the N-terminal and very C-terminal portion of the

Table 5
Summary of termination phenotype of Rif^r mutations

rpoB allele	Amino acid residue affected	Effect on terminator(s)† Simple	Rho-dependent	Suppression of *rho15*‡	Previously known termination phenotype§
Cluster I					
3445	Δ(507-511)	—	↑(1)	—	—
101	513	↑(1)	↑(3)	+	Suppressed *rho15*
8	513	↑(2)	↑(2)	—	Increased termination at *trp a* [0·2–0·5]
113	516	—	↑(1)	—	
148	516	—	↑(2)	—	
3051	517	—	↑(2)	—	
3595	522	↓(2)	↓(3)	—	
2	526	↓(1)	↓(2)	Inc.	Decreased termination at *trp a* [2·5–4·3]
3401	529	—	↓(3)	—	
3402	529	—	↓(2)	—	
114	531	—		—	
3449	Δ532	—	—	—	
3443	533	—		—	
Cluster II					
3370	563	↓(2)	↓(3)	+	
111	564	↓(2)	—	—	
7	572	↑(2)	↑(3)	—	Increased termination at *trp a* [0·3–0·6]
Cluster III					
3406	687	—	↓(2)	—	

† (↑) Increased or (↓) decreased in termination ≥1·5-fold by the termination assay described in the text. Number in parenthesis is the number of terminator(s) affected.
‡ (+) Suppressed or (−) not suppressed the termination defect caused by *rho15* mutation; Inc., incompatible with the *rho15* allele.
§ Number in brackets is the range of read-through of the mutant at several different terminators normalized to that of the *rpoB*⁺ strain (Yanofsky & Horn, 1981).

region increase termination while those in the central portion of the region decrease termination.

Almost half (6/14) of the Rif^r alleles altered termination at both simple and Rho-dependent terminators, suggesting that they affect processes common to both types of termination events. The

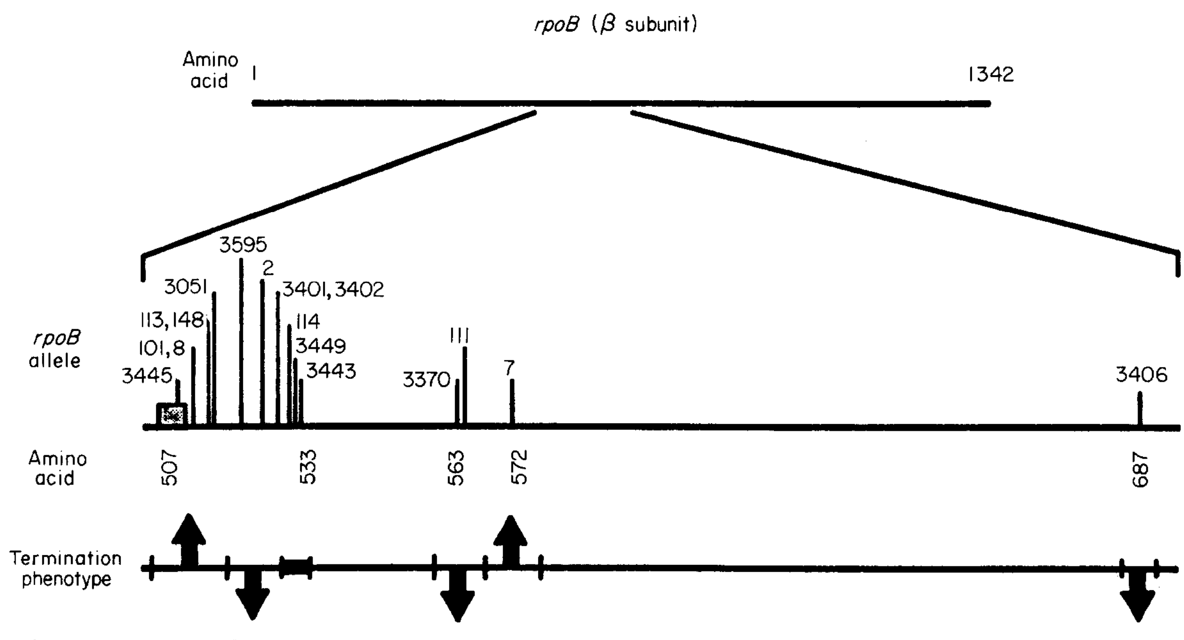


Figure 2. The location of Rif^r mutations and their termination phenotypes. A diagram of positions of Rif^r mutations in the β subunit. The horizontal line represents the β subunit. Vertical lines represent the positions of Rif^r mutations (the height of the lines has no significance). The box represents a Rif^r mutation that deletes 12 base-pairs affecting 5 amino acid residues. The effect of the mutations upon termination is indicated on the lower horizontal line. (↑) Signifies an increase in termination: (↓) signifies a decrease in termination; the filled box indicates no change in termination at the terminators assayed. The cluster numbers on the bottom line are referred to in the text.

fact that only a subset of terminators is affected by any one allele could occur if the termination efficiency is limited by different steps at different terminators. A particular Rif^r allele may alter one particular step in termination and only those terminators whose efficiency is determined by that step will show altered read-through.

The seven Rif^r alleles that altered termination only at Rho-dependent terminators could modify an interaction between RNA polymerase and Rho. However, the fact that only one of these mutants affected all three of the Rho-dependent terminators suggests that, if the Rif^r mutations do affect this interaction, it is dependent on terminator structure. We favor the idea that these mutations also affect terminator recognition. Their effects could be limited to Rho-dependent terminators because they alter a structural feature characteristic of Rho-dependent terminators only. Alternatively, the exclusive recognition of Rho-dependent terminators by these alleles could result from the set of terminators we used. Only two simple terminators were included in the set of terminator vectors. Assay of these Rif^r mutants on additional simple terminators might increase the number of mutants exhibiting altered read-through at this type of terminator.

Comparison of the Rif^r alleles increasing termination at Rho-dependent terminators with those suppressing the termination defect of *rho15* indicates that these two phenotypes are not correlated. Of the seven Rif^r alleles showing increased termination, only one, *rpoB101*, suppresses *rho15*. *rpoB3370*, the other Rif^r allele suppressing the termination defect of the *rho15* mutant strain, reduced termination at Rho-dependent terminators in the presence of wild-type *rho*. Thus, the Rif^r mutants do not restore termination by simply bypassing the need for Rho. Instead, the two suppressing RNA polymerases are likely to alter interaction either with the IS2 terminator or with a protein component of the transcription apparatus. The result of this altered interaction would be to restore *rho15* activity in termination. *In-vitro* transcription experiments support the idea that suppression involves restoration of Rho15 protein function. Rho15 protein exhibits ATPase and termination activity when transcription is carried out with RNA polymerase from the *rpoB101* mutant but is inactive with wild-type RNA polymerase (Das *et al.*, 1978).

The limited data in the literature provides support for the idea that Rif^r and wild-type RNA polymerases recognize terminators differently in a purified *in-vitro* transcription system. RNA polymerase from *rpoB8* (*rpoB203*) and *rpoB2* showed altered pausing and termination at the *trp* attenuator (Fisher & Yanofsky, 1983). These alterations were consistent with the *in-vivo* termination phenotypes of these strains at the *trp* attenuator. The ability of *rpoB203* (*rpoB8*) and wild-type RNA polymerase to terminate has been compared on a series of mutant simple terminators. RNA polymerase from *rpoB203* (*rpoB8*) mutants requires less extensive signals than does the wild-type to terminate at simple terminators. When a mutant terminator has either a very short G+C-rich hairpin loop or very few U residues, *rpoB203* (*rpoB8*) but not wild-type RNA polymerase can terminate efficiently *in vitro* (Christie *et al.*, 1981).

Why do mutations in RNA polymerase that alter rifampicin binding affect termination? Studies on the effects of rifampicin on transcription indicate that binding of rifampicin prevents elongation of RNA past a few nucleotides (Johnston & McClure, 1976; McClure & Cech, 1978) and also destabilizes the ternary complex (Schulz & Zillig, 1981). Furthermore, rifampicin binding is prevented when RNA as small as a trinucleotide is bound to the enzyme (Schulz & Zillig, 1981). These observations indicate that occupancy of the rifampicin binding site and the nucleotide binding site affect each other. It seems plausible, then, that mutations that alter the rifampicin binding site may directly or indirectly affect either nascent RNA or substrate binding. Alterations in either the length or stability of the nascent RNA–DNA hybrid could affect the transcription bubble and the likelihood of transcript release at pause sites, while altered substrate binding could change the elongation rate and also affect pausing and termination at particular terminators. *In-vitro* analysis of the Rif^r RNA polymerases should help determine whether such processes are affected by mutation to rifampicin resistance and if these effects are correlated with termination efficiency at particular terminators.

We thank Drs K.-C. Luk, W. Szybalski and M. Chamberlin for providing plasmids. We also thank Dr S. Adhya for helpful advice on strain construction with the *rho15* allele. Finally we thank our colleagues Deborah Cowing, Alan Grossman, Will McClure, Terry Platt and David Straus for helpful comments on the manuscript. This work was supported by a grant from the National Institutes of Health to C.A.G.

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Edited by M. Gottesman